

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

OPN403

Scaling Argo CD efficiently to handle thousands of applications

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Agenda

What is Argo CD and why should we care?

Our approach

Summary of scalability testing and tuning work

Sharding deep dive

Conclusions and future activities

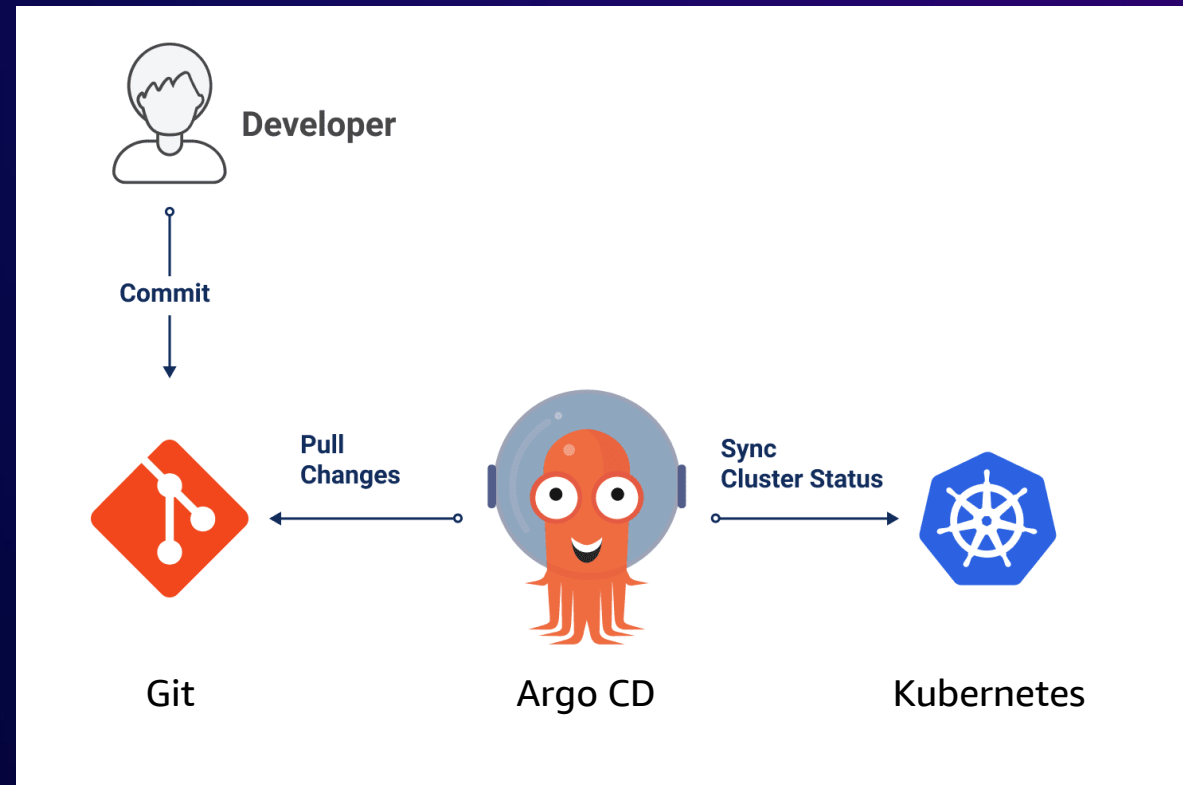
Chalk talk

What is Argo CD and why do we care?

Popular open sourced GitOps deployment solution

Large-scale organizations are increasingly adopting Argo CD

Not a lot of scalability testing has been done in the community, and implications of using Argo CD at scale are not known



Our approach

- Provide tooling and resources to perform scalability testing of Argo CD
- Identify existing performance settings and parameters
- Present scalability testing results to the community and pinpoint performance bottlenecks
- Work with the community to contribute fixes that solve for identified bottlenecks

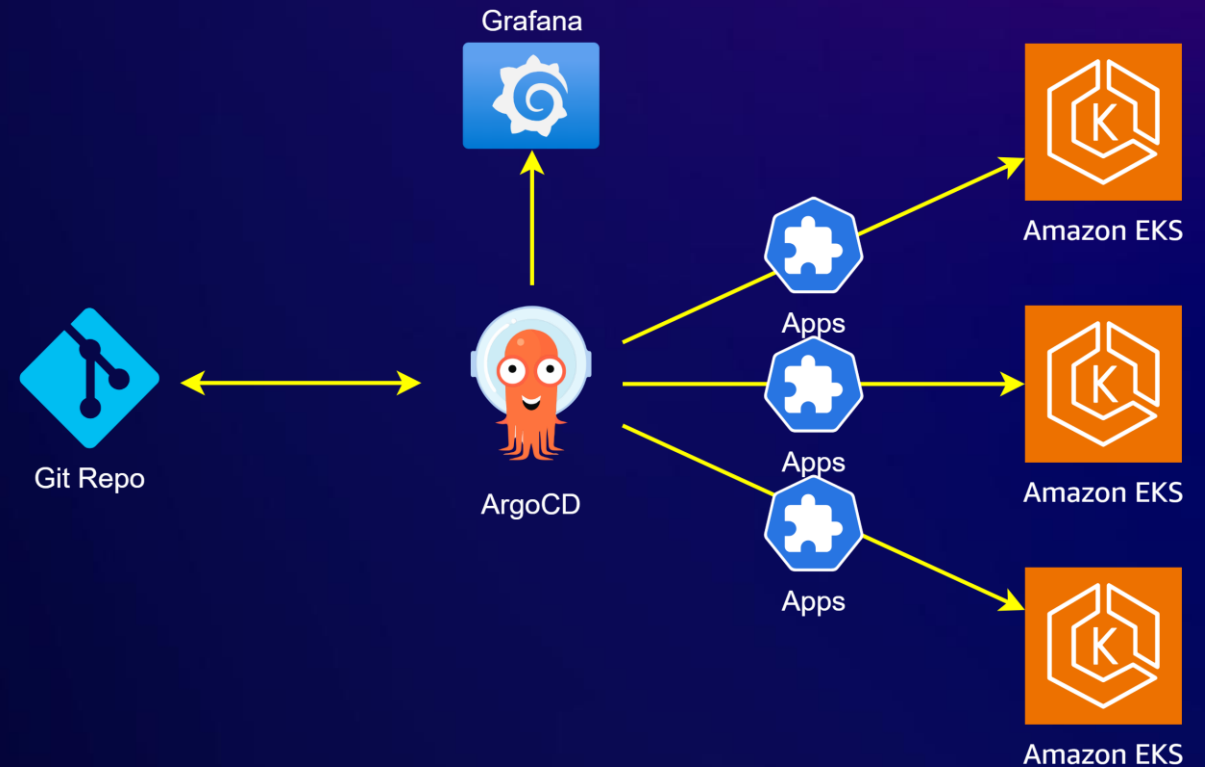
Summary of scalability testing and tuning work

First round documented in
blog post

- Key metrics
- Key scalability parameters

Second round of testing

- **Applications: 15,000, 30,000, 50,000**
- **Clusters: 100, 250, 500**
- **Amazon EKS Nodes: 1, 2, 3**



Sharding deep dive

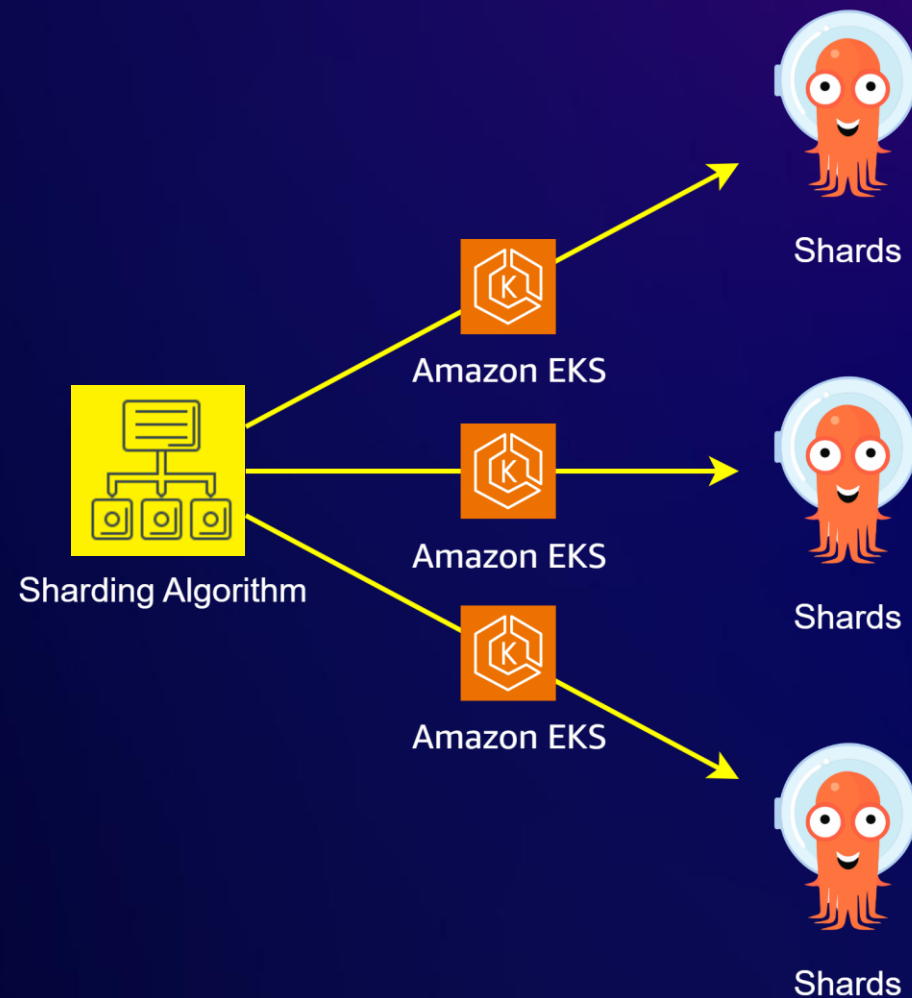
How sharding works

Sharding by cluster

- Legacy
- Round-robin

Sharding by apps (future)

- Greedy minimum
- Weighted ring hash
- Bounded loads



Sharding deep dive

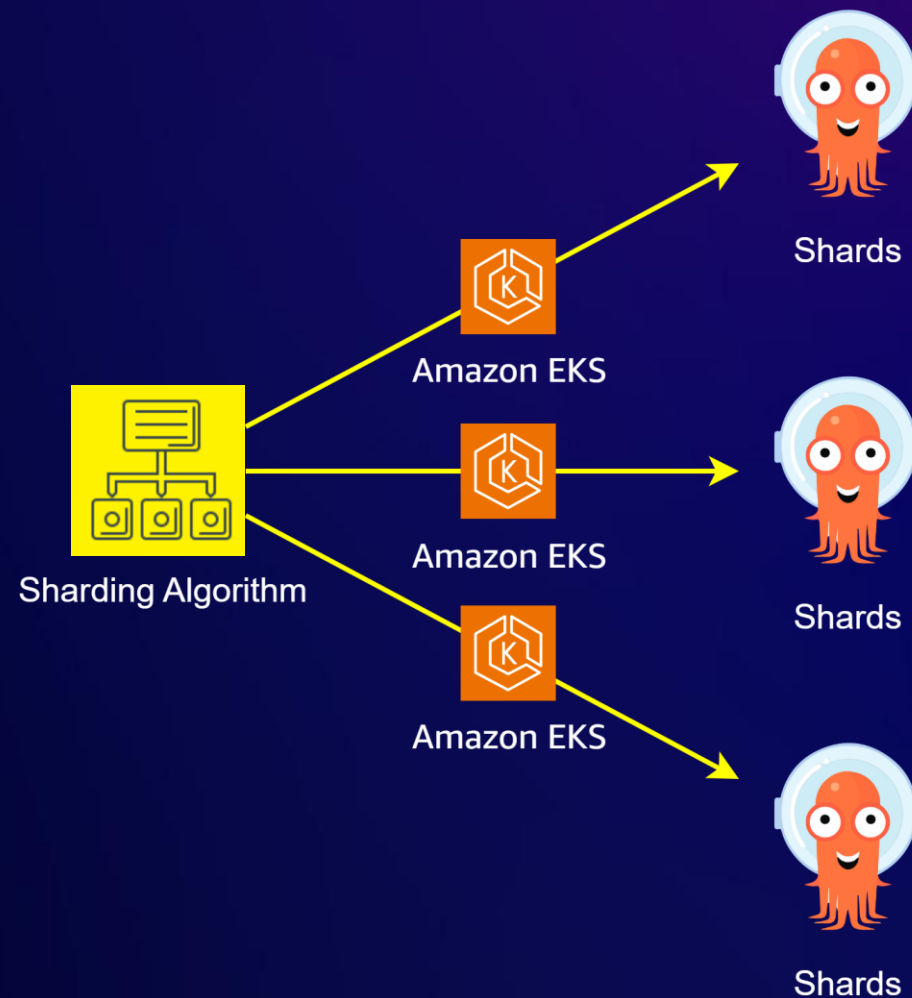
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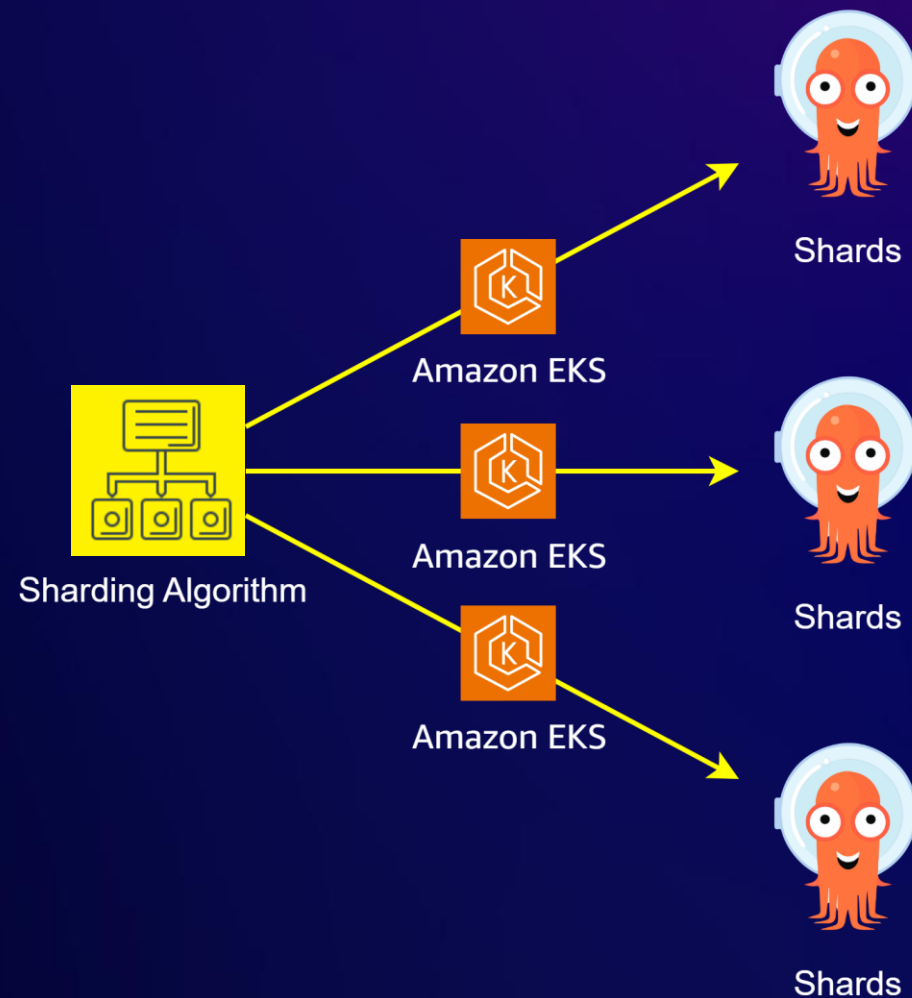
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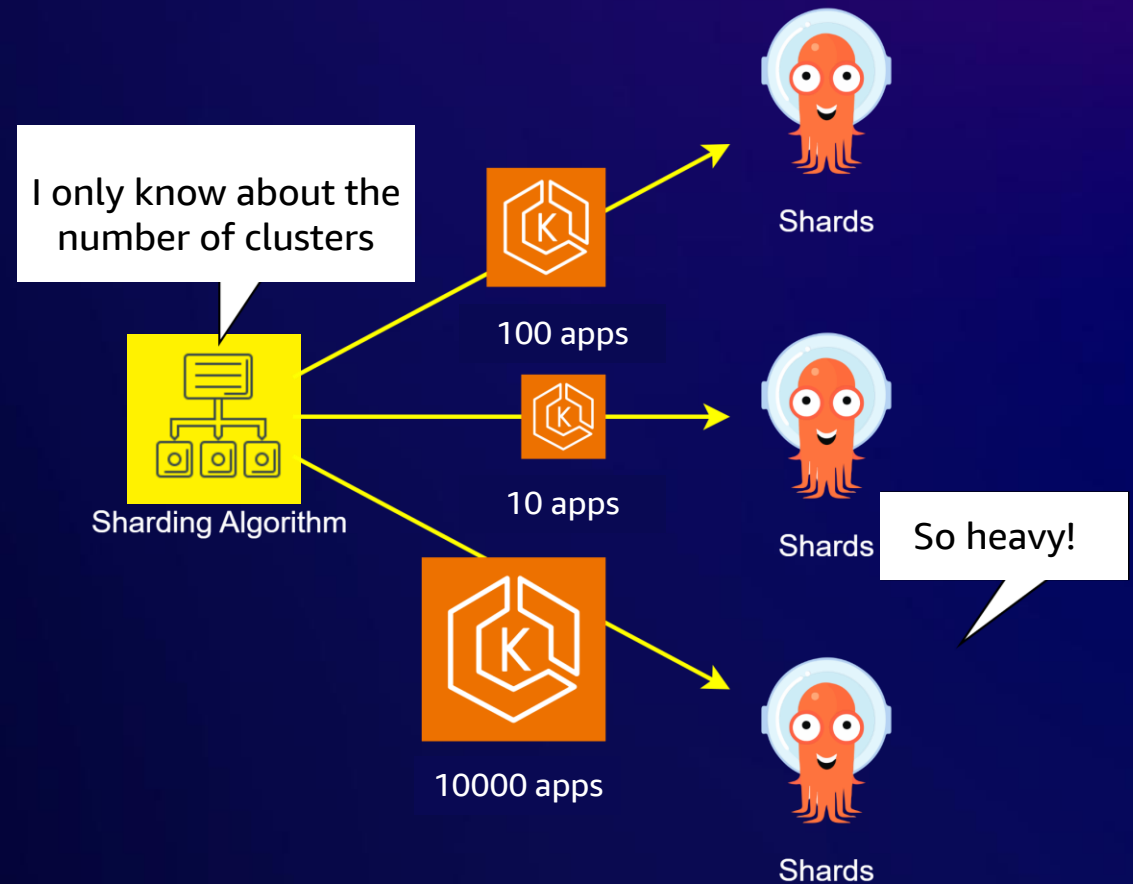
Sharding by cluster

Legacy

- Simple
- Does not maintain balance across shards (range: 0.32)

Round-robin

- Balance across shards (range: 0.02)
- Only works when number of applications is equal across clusters
- Balance when randomizing apps (range: 0.27)
- Rebalancing issues!



Sharding by apps (newly developed)

Greedy minimum

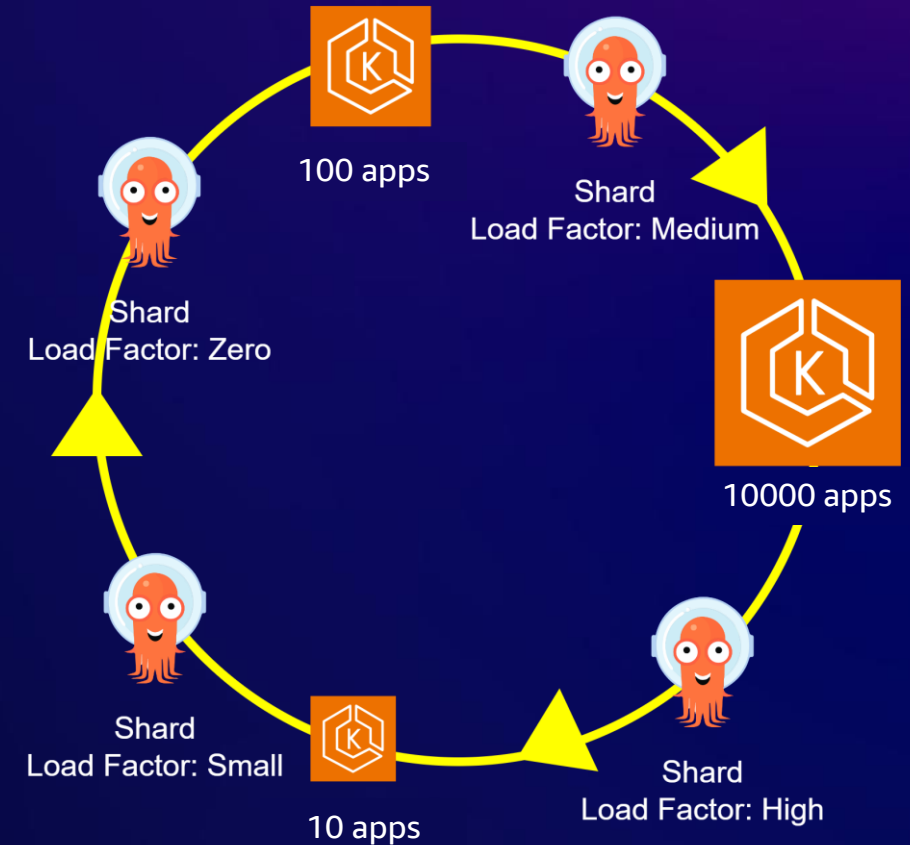
- Keeps shards balanced (range: 0.06)
- Rebalancing issues: 75% of clusters reassigned

Weighted ring hash

- Decreased shard balance (range: 0.12)
- Better rebalancing: 48% of clusters reassigned

Bounded loads

- Slightly decreased shard balance (range: 0.17)
- Best rebalancing: Only 17% of clusters reassigned



Conclusions and future activities

- Sharding is a key component to scaling Argo CD
- Implement new sharding algorithms
- Explore re-architecting app controller

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Thank you!

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